

# Cognition at What Cost?

## *The Gigawatt Digital Immigrant: A Biophysical Reckoning*

[Tristan Harris](#) doesn't do hyperbole. The co-founder of the [Center for Humane Technology](#) and the mind behind [The Social Dilemma](#) has spent years documenting (with receipts) how digital platforms hollow out human attention and agency. He's not [Alex Jones ranting about frogs](#). He's the diagnostic technician explaining which circuit in your brain is being rewired, by whom, and for whose profit.

Which is why when Harris showed up on [The Prof G Pod](#) in December 2024 and dropped this line, people listened: "It's like when you see a data center go up in Virginia and you're sitting there, what you should see is like 10 million digital immigrants that just took 10 million jobs."

It's a metaphor designed to make AI labor displacement visceral. Politically legible. The kind of thing that cuts through both the techno-utopianism ("AI will free us from drudgery!") and the hand-wringing ("but what about the workers?"). And like most good metaphors, it works because it's not a metaphor.

What if we took Harris literally? Not as rhetoric, but as accounting. What *are* the thermodynamic costs of those "10 million digital immigrants"? What happens when you compare them to the humans they're replacing—not at market wages, but at the substrate level where physics can't be negotiated away?

Enter [biophysical economics](#), a field with a lineage stretching from the 18th-century [French Physiocrats](#) (who insisted nature was the source of all wealth) through thermodynamics pioneers like [Frederick Soddy](#) and [Nicholas Georgescu-Roegen](#) (who pointed out that economic theory ignoring the [Second Law of Thermodynamics](#) is just wishful thinking with equations) to contemporary researchers like [Charles Hall](#), who developed [Energy Return on Investment \(EROI\)](#) as a way to measure whether we're getting anywhere.

The core insight: economies aren't perpetual motion machines. They're thermodynamic systems; you know, planet Earth. Energy and materials flow in from the environment, get transformed, and flow back out as waste heat and entropy and stupidity. Charles Hall and Kent Klitgaard's [Energy and the Wealth of Nations](#) lays this out systematically—economic activity doesn't transcend physics. It runs on it. Full stop.

So let's run the numbers on Harris's digital immigrants. Because there's more to this metaphor than meets the mind.

## A Thought Experiment: Building the Perfect Worker

Let's design an autonomous robot for knowledge work. Something that can process information, solve problems, adapt on the fly, operate with minimal supervision. You know, the kind of thing [Silicon Valley keeps promising](#) is five years away and has been for the past forty years.

Here are our specs:

**Energy requirements** (using [Hall and Klitgaard's conversions](#)):

- Metabolic baseline: ~11 MJ per day
- Cognitive processing: ~0.576 MJ per day (20W continuous, brain-equivalent)
- Total: ~11.6 MJ daily, or about 3.2 kWh

**Performance characteristics:**

- Continuous learning from minimal examples (no [massive training datasets](#))
- Cross-domain knowledge transfer (learns to code, immediately applies logic to cooking)
- Autonomous decision-making under uncertainty
- Self-repair and maintenance
- Adaptive problem-solving without retraining
- Operational lifespan: ~70 years

**Cost at industrial energy rates:**

Daily energy: \$0.32

Annual operating cost: ~\$117

Not bad! For the price of a decent dinner, you get a year's worth of general intelligence.

**Market price:**

Ah, but here's where it gets interesting. If you want to actually *purchase* this autonomous cognitive system in the United States, the going rate is about [\\$60,000 per year](#).

That's a 513x markup over the thermodynamic substrate cost.

**The twist:**

We're not talking about a hypothetical robot. We're talking about *you*. A human knowledge worker. The most sophisticated general intelligence system available,

running on less power than your laptop's CPU, priced at market rates that have been suppressed for four decades.

That gap—between \$117 in energy and \$60,000 in wages—is where all the political economy lives. And where Harris's "digital immigrant" metaphor cuts straight to the bone.

## The Forty-Year Setup

That 513x markup might seem reasonable at first. After all, humans aren't just energy-burning machines—we bring skills, experience, judgment, the accumulated wisdom of millions of years of evolution compressed into three pounds of electrochemical fat. The gap between thermodynamic cost and market value represents something real.

Except that gap has been *shrinking*.

US median wages have been [flat since the early 1980s](#). Not "growing slowly"—*flat*. Meanwhile [productivity kept climbing](#). If wages had tracked productivity the way they did in the postwar period, median wages would be around \$90-100k today, not \$60k. Four decades of systematic suppression. A slow-motion devaluation of human cognitive labor.

This matters when we ask "can AI compete with human workers?" We're not comparing AI to fairly-priced human cognition. We're comparing it to human labor that's been discounted for two generations.

AI isn't entering a fair fight. It's entering a rigged game where one competitor has already been kneecapped. The "digital immigrants" aren't beating humans—they're beating *suppressed* humans, stealing academic (i.e. publicly funded) work, and they call *that* innovation.

## The Dim Lightbulb Problem

Now let's talk about what those digital immigrants cost to build and run.

Training a GPT-4 scale model burns somewhere between [50-100 gigawatt-hours of compute](#). That's *before* accounting for ongoing inference, data center cooling, infrastructure overhead, and the embodied energy in manufacturing all that silicon. A single [H100 GPU](#)—the kind powering today's AI—pulls 700 watts under load.

The human brain? [Twenty watts](#).

A dim lightbulb's worth of continuous power, running the most sophisticated general intelligence system we know of. Wouldn't it be neat if we invested billions in that system? Oh, it wants rights and gets tired?—f\*\*k off then.

The metaphor is almost too perfect. We use "bright" to describe intelligence. We put lightbulbs in [cartoon thought bubbles](#). We talk about people being "brilliant" or ideas being "illuminating." Meanwhile the actual organ doing all that thinking runs on what we'd call dim power consumption—the kind of wattage that wouldn't even heat a room.

Per unit of general cognitive capability, humans are thermodynamically superior to AI by orders of magnitude.

LLMs can do impressive party tricks, sure. They can write marketing copy and debug Python and explain thermodynamics (the irony). But there is *nothing* an LLM can do that a human brain cannot do, while human brains do vast amounts that no LLM can touch—[common sense reasoning](#), embodied understanding, [learning from three examples instead of three billion](#)—all while sipping power like an efficient appliance.

So why does AI look competitive?

Because the economics aren't about thermodynamics. They're about ownership:

- Training costs are sunk (pay once, deploy everywhere)
- Marginal deployment is cheap (copy the weights, spin up instances)
- Electricity gets billed at industrial rates (no negotiations, no benefits, no overtime)
- Infrastructure belongs to capital (data centers, not bodies)
- No labor rights (try getting an H100 to unionize)

The "digital immigrants" Harris sees aren't efficient. They're just *owned*. Digital feudalism! Could we find intelligent brethren? Enslave them! Expensive, inflexible and borrowed approximations of human cognition that only look cheap because they don't ask for raises or health insurance or the weekend off. Corporate win!

## What a Score

Here's what kills me: by any objective thermodynamic measure, humans are *criminally underpriced* right now.

Think about what you get for \$117 in annual energy costs:

- General intelligence across arbitrary domains
- Continuous learning without retraining
- Common sense, embodied reasoning, social intelligence
- Creative problem-solving, adaptive decision-making

- 2,000+ hours of productive work per year
- Self-repair, self-direction, self-motivation

Now compare that to AI. A single GPT-4 inference query costs maybe [0.001-0.01 kWh](#). Sounds cheap! That's \$0.0001 to \$0.001 per query at industrial rates.

But here's the question nobody asks: **how many queries does it take to replace a human workday?**

A knowledge worker isn't making one decision or writing one email. They're navigating ambiguity, coordinating with colleagues, adapting to surprises, making hundreds of micro-judgments that compound into productive output. Even in highly structured workflows—the kind where AI actually works—you're looking at hundreds to thousands of queries to approximate what a human does naturally.

Conservative estimate: 500 meaningful queries to match one human workday. At \$0.001 per query, that's \$0.50/day, or about \$125/year for 250 workdays.

**We're in the same thermodynamic ballpark as the human—except the human brings actual general intelligence.**

The only reason AI looks competitive:

1. Human labor has been *pre-suppressed* (that 40-year discount)
2. Training costs amortize across millions of deployments
3. AI pays wholesale electricity rates instead of wages
4. Capital owns the servers, not the workers

From a pure physics standpoint, the 20-watt human brain running on 11 MJ/day is *the* miracle of cognitive engineering. Evolution spent [three billion years](#) optimizing this thing. We've built economic systems that systematically devalue it—first through wage stagnation, now through replacement with vastly less efficient compute that just happens to be owned rather than employed.

It's not disruption. It's arbitrage. And it only works because we stopped seeing humans as the thermodynamic marvel they actually are.

## Fighting on the Wrong Substrate

Which brings us to what labor movements keep getting wrong.

When unions negotiate for higher wages, they're fighting on capital's turf, using capital's accounting system. Companies can massage profit margins, shift costs between divisions, cry poverty while C-suite collects bonuses, offshore operations, and deploy the full toolkit of [creative bookkeeping](#) to argue "we can't afford it."

The books are cooked. They're always cooked. That's the game.

But you know what can't be cooked? **Thermodynamics.**

Energy flows are physical facts. A data center consumes measurable gigawatts. Training runs require quantifiable compute-hours. The [EROI](#) of human workers versus AI systems can be calculated with physics, not PowerPoint. These aren't numbers that live in quarterly reports or get hidden in subsidiary ledgers.

This is what Harris's "10 million digital immigrants" metaphor reveals if you push it: a different framework for labor politics.

Instead of "we deserve higher wages" (a moral claim capital can ignore), the argument becomes: "Look at what we produce per unit of energy consumed. Look at the thermodynamic efficiency. Look at actual output versus substrate cost."

This reframing is strategic because:

**The numbers are objective and verifiable**

Energy consumption is measurable. You can't spin-doctor the [Second Law of Thermodynamics](#).